Chapter 8: Heaps

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 8 (Heaps) of Data Structures.

Technical / Interview Questions

1. What is a heap data structure? Explain its fundamental properties.
2. Define the heap property and explain how it is maintained.
3. Distinguish between Min-Heap and Max-Heap with examples.
4. Explain the array representation of a binary heap.
5. Describe the insertion operation in a binary heap.
6. Explain the Extract-Min operation in a binary heap.
7. What is Heapify? Why is it required in heap operations?
8. Explain the structure and properties of binomial trees.
9. What is a Binomial Heap? How does it differ from a binary heap?
10. Explain the union operation in Binomial Heaps.
11. Describe the insert operation in Binomial Heaps.
12. Explain how Extract-Min works in Binomial Heaps.
13. What is a Fibonacci Heap? Mention its important characteristics.
14. Explain the concept of lazy operations in Fibonacci Heaps.
15. What is the Decrease-Key operation in Fibonacci Heaps?
16. Explain the cascading cut mechanism in Fibonacci Heaps.
17. Describe the consolidation process during Extract-Min in Fibonacci Heaps.
18. Compare Binary Heap and Binomial Heap.
19. Compare Binomial Heap and Fibonacci Heap.
20. What are the advantages of Fibonacci Heaps over binary heaps?
21. Why are Fibonacci Heaps useful in graph algorithms?
22. Explain how heap data structures are used in priority queues.
23. Discuss the time complexity of heap operations in different heap structures.
24. What are the applications of heaps in operating systems?
25. Explain how heaps are used in scheduling algorithms.

Programming Exercises

1. Write a program to insert an element into a binary heap.
2. Write a program to delete the root element from a binary heap.
3. Implement the Heapify procedure for a binary heap.
4. Write a program to build a binary heap from an unsorted array.
5. Implement the heap sort algorithm.
6. Write a program to find the kth largest element using a heap.
7. Implement a priority queue using a binary heap.
8. Write a program to merge two binary heaps.
9. Implement insert operation in a Binomial Heap.
10. Write a program to perform union operation in a Binomial Heap.
11. Implement extract-min operation in a Binomial Heap.

12. Write a program to decrease key in a Fibonacci Heap.
13. Implement Extract-Min operation in a Fibonacci Heap.
14. Write a program to delete a node from a Fibonacci Heap.
15. Implement a priority queue using Fibonacci Heap operations

Multiple Choice Questions

1. A heap is primarily used to implement:

(a) Stack
(b) Queue
(c) Priority Queue
(d) Graph

2. In a min-heap, the smallest element is located at:

(a) Leaf node
(b) Root node
(c) Middle node
(d) Any node

3. In a max-heap, the parent node is always:

(a) Smaller than children
(b) Equal to children
(c) Greater than or equal to children
(d) Random

4. The time complexity of inserting an element into a binary heap is:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) O(n log n)

5. The extract-min operation in a binary heap takes:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) $O(n^2)$

6. The heap sort algorithm has time complexity:

(a) O(n)
(b) O(n log n)
(c) O(log n)
(d) $O(n^2)$

7. A binary heap is typically implemented using:

(a) Linked list
(b) Array
(c) Stack
(d) Hash table

8. A Binomial Heap is composed of:

(a) Binary trees
(b) AVL trees
(c) Binomial trees
(d) B-trees

9. The union operation of two Binomial Heaps takes:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) O(n log n)

10. In Fibonacci Heaps, most operations run in:

(a) O(n)
(b) O(log n)
(c) O(1) amortised
(d) O(n log n)

11. Which heap supports O(1) amortised decrease-key operation?

(a) Binary heap
(b) Binomial Heap
(c) Fibonacci Heap
(d) AVL tree

12. Cascading cuts are used in:

(a) Binary heap
(b) Binomial Heap
(c) Fibonacci Heap
(d) Red-black tree

13. Fibonacci Heaps are especially useful in:

(a) Sorting algorithms
(b) Graph algorithms
(c) Searching algorithms
(d) String matching algorithms

14. Which algorithm benefits from Fibonacci Heaps?

(a) Bubble sort
(b) Dijkstra's algorithm
(c) Linear search
(d) Selection sort

15. Heap sort is an example of:

(a) Comparison-based sorting
(b) Non-comparison sorting
(c) Hash-based sorting
(d) Graph sorting

16. The maximum number of children in a binomial tree of order k is:

(a) k
(b) 2k
(c) k^2
(d) log k

17. In a binary heap stored in an array, the left child index of node i is:

(a) i/2
(b) 2i
(c) 2i+1
(d) i+1

18. The right child index of node i in an array heap is:

(a) 2i+1
(b) 2i
(c) i/2
(d) i−1

19. The parent of node i in a binary heap array is:

(a) i/2
(b) 2i
(c) i+2
(d) 2i+1

20. The build heap operation takes:

(a) O(n)
(b) O(log n)
(c) O(n log n)
(d) $O(n^2)$

21. In a Fibonacci Heap, consolidation occurs during:

(a) Insert
(b) Union
(c) Extract-Min
(d) Find-Min

22. Which heap structure supports efficient merge operation?

(a) Binary heap
(b) Binomial Heap
(c) Fibonacci Heap
(d) Both B and C

23. The degree of a binomial tree Bk is:

(a) k
(b) k+1
(c) 2k
(d) log k

24. The number of nodes in a binomial tree Bk is:

(a) k
(b) 2k
(c) 2^k
(d) k^2

25. A priority queue implemented with a binary heap supports insert in:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) O(n log n)

26. Which heap allows constant-time merge operation?

(a) Binary heap
(b) Binomial Heap
(c) Fibonacci Heap
(d) AVL tree

27. Heap data structures are commonly used in:

(a) CPU scheduling
(b) Network routing
(c) Graph algorithms
(d) All of the above

28. In heap sort, the heap structure used is usually:

(a) Min heap
(b) Max heap
(c) Binomial Heap
(d) Fibonacci Heap

29. Fibonacci Heaps were introduced by:

(a) Donald Knuth
(b) Fredman and Tarjan
(c) Dijkstra
(d) Hoare

30. The main advantage of Fibonacci Heaps is:

(a) Simplicity
(b) Faster decrease-key operations
(c) Smaller memory usage
(d) Easier implementation

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	C	4	B	5	B
6	B	7	B	8	C	9	B	10	C
11	C	12	C	13	B	14	B	15	A
16	A	17	B	18	A	19	A	20	A
21	C	22	D	23	A	24	C	25	B
26	C	27	D	28	B	29	B	30	B